

Yunhan Qiao

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Ph.D. Candidate, Computer Science, Oregon State University

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RESEARCH INTERESTS

My research sits at the intersection of empirical software engineering, code comprehension, and human-AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

EDUCATION

- **Ph.D. in Computer Science, School of EECS** Oregon State University, Corvallis, OR
Ph.D. Candidate
Sep 2023 – Jun 2027 (*expected*)
 - **Advisor:** Dr. Christopher Hundhausen at the [Software Engineering Education Assessment and Analytics \(SEAL\) Lab](#)
 - **Research Areas:** Empirical Software Engineering, Code Comprehension, Computing Education, Human-AI Interaction
- **M.S. in Software Engineering** Arizona State University, Tempe, AZ
Thesis track
Aug 2021 – May 2023
 - **Advisor:** Dr. Robert LiKamWa
 - **Thesis:** Networked System for Volumetric Athletic Coaching in Augmented Reality
- **B.E. in Software Engineering** Guangxi University of Science and Technology, Liuzhou, China
Dual degree program with Southern Cross University
Sep 2016 – Jun 2020
- **B.E. in Information Technology** Southern Cross University, Gold Coast, Australia
Dual degree program with Guangxi University of Science and Technology
Sep 2017 – Jun 2020

PUBLICATIONS

Journal Articles & Conference Papers

- Yunhan Qiao, Md Istiak Hossain Shihab, and Christopher Hundhausen. **A Systematic Literature Review of the Use of GenAI Assistants for Code Comprehension: Implications for Computing Education Research and Practice.** *TOCE: ACM Transactions on Computing Education*, 26(2), Article 24, 1–33, 2026. [\[PDF\]](#) [\[DOI\]](#)
- Ahsun Tariq, Phillip Conrad, Christopher Hundhausen, S. Chandrasekaran, Yunhan Qiao, and Summit Haque. **Using Slackbot-Prompted Reflections to Promote Participation and Reduce Contribution Inequalities in Team Software Development Projects.** *TOCE: ACM Transactions on Computing Education*, 2026.
- Md Istiak Hossain Shihab, Christopher Hundhausen, Ahsun Tariq, Summit Haque, Yunhan Qiao, and Brian Wise Mulanda. **The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Coding Tasks.** *ICER'25: Proceedings of the 2025 ACM Conference on International Computing Education Research*, Vol. 1, 407–420, Aug. 2025. [\[PDF\]](#) [\[DOI\]](#)
- Manish Motwani, Aakash Kulkarni, Yunhan Qiao, Matthew Davis, and Ziyang Chen. **LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications.** *AIxSE'25: IEEE International Conference on AI x Software Engineering*, 2025. [\[PDF\]](#)

Under Review

- Yunhan Qiao, Md Istiak Hossain Shihab, Summit Haque, and Christopher Hundhausen. **Faster but Not Wiser: GitHub Copilot Decouples Programming Performance from Code Comprehension in Brownfield Tasks.** *TOCE: ACM Transactions on Computing Education*, under review. [\[arXiv\]](#)
- Yunhan Qiao, Summit Haque, and Christopher Hundhausen. **Beyond the Prompt: Linking What Developers Ask, Do, and Understand with Coding Agents.** *SANER'27: IEEE International Conference on Software Analysis, Evolution and Reengineering*, under review. [\[arXiv\]](#)

Under Submission

- O. Schutz, Yunhan Qiao, F. Moussaoui, R. Choudhuri, and A. Sarma. **Heart and Mind in the Loop: Why We Vibe When We Work with AI.** *CHASE'27: International Conference on Cooperative and Human Aspects of Software Engineering*, under submission.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant, School of EECS** Oregon State University
Undergraduate software engineering courses
2024 – current
 - **CS 362:** Software Engineering II
 - **CS 461:** Senior Software Engineering Project
 - **CS 467:** Online Capstone Project
 - **CS 492:** Mobile Software Development

PROFESSIONAL EXPERIENCE

- **Software Engineering Intern, Mocha Software**
Software development

Nanjing, China
Aug 2020 – Nov 2020

PROFESSIONAL SERVICE

- **Program Committee:** SIGCSE TS 2027, the 58th ACM Technical Symposium on Computer Science Education

SKILLS & COURSEWORK

- **Research Methods:** Controlled experiments, mixed methods, EEG studies, behavioral & log analysis, prompt coding, interviews, thematic analysis, survey design (Qualtrics), statistical analysis
- **Programming:** Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
- **Frameworks & Tools:** React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, L^AT_EX; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
- **Coursework:** Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis
- **Languages:** English (fluent), Mandarin Chinese (native)